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SCHOOL MANAGEMENT SYSTEM

UML CLASS DIAGRAM REPORT (REPORT 2)

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1. Classes, Attributes, and Operations (Methods)

- Student Class:
  * Attributes:
    - studentId: int
    - fullName: String
    - email: String
    - dateOfBirth: Date
  * Methods (Operations):
    - registerCourse(): void
    - viewGrades(): void

- Teacher Class:
  * Attributes:
    - teacherId: int
    - fullName: String
    - specialization: String
    - email: String
  * Methods (Operations):
    - assignGrade(): void
    - viewSchedule(): void

- Course Class:
  * Attributes:

- courseId: int

- courseName: String

- credits: int

* Methods (Operations):

- updateDetails(): void

- Enrollment Class:

* Attributes:

- enrollmentId: int

- enrollmentDate: Date

- status: String

* Methods (Operations):

- confirmEnrollment(): void

- cancelEnrollment(): void

2. Relationships and Multiplicities (Cardinalities)

- Student to Enrollment (1 to 0..*):

* Explanation: A single student (1) can have multiple enrollment records (0..*) across different terms or courses.

- Course to Enrollment (1 to 0..*):

* Explanation: A single course (1) can accommodate multiple student registrations (0..*) through enrollment records.

- Teacher to Course (1 to 1..*):

* Explanation: A single teacher (1) can teach one or more courses (1..*) in the curriculum.

3. Summary of System Structure

The static structure demonstrates a clean relational pipeline where students enroll in courses via the Enrollment association class, while teachers manage the courses they instruct.